

CITY TWIN

Team Brief · EuroTech × HKTE Hackathon · Munich

The human integration infrastructure every global city needs.

Hong Kong has the opportunities. City Twin connects people to them.

A city is not truly smart if the people living inside it feel lost.

1. WHAT IS CITY TWIN

City Twin is the AI layer that makes cities navigable for people — connecting each person, in their own language, with the opportunities and communities they need to thrive.

CITY TWIN IS

- The AI layer that makes cities navigable for humans
- The retention infrastructure HK is missing
- The first person who greets you in a new city and tells you exactly where to go

2. THE PROBLEM

Every year, thousands of young internationals arrive in Hong Kong full of talent and ambition. They have exactly what this city needs. But 40% leave within the first year. Not because of a lack of possibilities — Hong Kong has thousands. But because they can't find them and because they don't feel connected.

Pain 1 — Opportunity overload: opportunities are invisible

HK has Cyberport, HKSTP, 200+ accelerators, government grants, university scholarships, talent visa programmes. All of it exists. But the newcomer doesn't know if they qualify, and has no one to explain it. Opportunities are lost before anyone uses them.

Pain 2 — Social isolation: the silent killer

40% of expats leave HK in the first year. Not because of work — because of loneliness. Because they can't find their community. Because they feel foreign inside a city that has exactly what they need. Social isolation is HK's biggest retention risk and the least measured.

Pain 3 — Language barrier: the invisible wall

Most opportunity platforms are in English only. Talent arriving from mainland China, France, Spain, Latin America, India — faces an additional barrier that shouldn't exist in 2026. City Twin eliminates it completely: speak your language, HK understands you.

The consequence — HK loses talent it already has

Hong Kong invests millions in talent attraction programmes: HKTE, Quality Migrant Admission Scheme, Top Talent Pass Scheme. But once talent arrives, the system abandons them. There is no integration infrastructure. City Twin is exactly that missing piece.

3. TWO USERS. TWO PAINS. ONE SOLUTION.

City Twin serves two distinct users — and HKTE needs both.

USER A — ALREADY IN HONG KONG

- Arrived with ambition. Now lost and alone.
- Has a job but no social life.
- Does not know what opportunities exist.
- Risk: leaves HK within months.

USER B — NOT YET IN HONG KONG

- Wants to go but doesn't know what to expect.
- Unsure if there are opportunities for their profile.
- Considering Singapore or Dubai instead.
- Risk: never arrives. HK loses them to a competitor city.

City Twin solves both. User A needs community, orientation and integration. User B needs validation, a pre-arrival map and proof that HK is the right move. HKTE has two mandates — attract talent AND retain it. City Twin is the only tool that does both at once.

4. THE THREE DEMO PROFILES

These are the three real people City Twin helps. Each one represents a different pain, a different language, and a different moment in their Hong Kong journey. These are the profiles used in the live demo on stage.

Sofia — Student, Madrid

Spanish · Considering Hong Kong

Has not arrived yet. Exploring whether HK is right for her.

PROFILE TAGS: Scholarships & funding · Spanish community · Volunteering

Her situation:

"I'm a university student in Madrid thinking about going to Hong Kong. I want to know what scholarships exist for European students, whether there's a Spanish-speaking community there, and I'd love to do some volunteering. I have no idea where to start."

City Twin shows her:

HKU and HKUST scholarships for European students up to HKD 120K/year. The Spanish Chamber of Commerce HK with 3,000+ members. Erasmus Alumni HK network. HKTE Volunteer Programme. Foodlink HK food bank network. Student Union and CUHK International House for campus life.

Constellation nodes generated:

HKU Scholarship · HKUST Global · Spanish Chamber HK · Erasmus Alumni HK · EuroExpats HK · HKTE Volunteer · Foodlink HK · Student Union HKU · CUHK Int'l House

Lena — Entrepreneur, Berlin

German · 1 month in Hong Kong

Already arrived. Looking for funding and her founding community.

PROFILE TAGS: Startup funding · AI community HK · International founders

Her situation:

"Ich bin seit einem Monat in Hongkong. Ich habe eine Idee im Bereich KI und suche nach Finanzierung. Ich kenne kaum jemanden hier und möchte andere internationale Gründer treffen, besonders in der AI-Szene." (I have been in Hong Kong for a month. I have an AI idea and I am looking for funding. I barely know anyone here and want to meet other international founders, especially in the AI scene.)

City Twin shows her:

Cyberport Incubation Programme (up to HKD 500K + free office space). HKSTP Incubation for deep tech. Zeroth.AI pre-seed accelerator focused on AI startups. Brinc Accelerator with AI & IoT track. AI Community HK for weekly events and co-founder matching. StartupHK Slack with 8,000+ founders. Women Founders HK mentorship circles. German Chamber of Commerce HK for her national network.

Constellation nodes generated:

Cyberport · HKSTP Incubation · InvestHK · Brinc Accelerator · Zeroth.AI · AI Community HK · StartupHK · Women Founders HK · German Chamber HK · RISE Conference



Priya — AI Researcher, India

Indian · Just arrived in Hong Kong

Just landed. Wants to start a company. Doesn't know where to begin.

PROFILE TAGS: AI research · Startup path · Find co-founders

Her situation:

"I am an AI researcher, just graduated from India. I want to start a company in Hong Kong but I don't know where to begin. I don't know anyone here and I'm not sure if my research background is relevant for building a startup."

City Twin shows her:

HKSTP AI Lab — research-to-market programme, directly relevant to her background. InnoHK Research Clusters with 28 R&D centres. Cyberport Incubation for early-stage funding. Zeroth.AI pre-seed accelerator for AI founders. AI Community HK for co-founder matching. Indian Founders HK — a growing community of Indian entrepreneurs. HKTE Platform for visa and talent scheme guidance. Free Cantonese basics course through HKTE.

Constellation nodes generated:

HKSTP AI Lab · InnoHK Cluster · Cyberport · InvestHK · Zeroth.AI · AI Community HK · StartupHK · Indian Founders HK · HKTE Platform · Cantonese Basics

5. WHAT USERS FIND INSIDE CITY TWIN

Specific and real.

FUNDING - Cyberport Incubation (up to HKD 500K) - HKSTP programmes - Innovation & Technology Fund (ITF) - InvestHK free advisory - Brinc, Zeroth.AI, Blueprint (HSBC)	SCHOLARSHIPS & EDUCATION - HKU, HKUST, CUHK scholarships - HKTE talent admission schemes - Research grants (RGC) - Exchange programmes - Free Cantonese courses (HKTE)
COMMUNITIES BY NATIONALITY Spanish Chamber of Commerce HK	ENTREPRENEURSHIP COMMUNITIES StartupHK (8,000+ members)

• French Chamber HK • German Business Association HK • Indian Founders HK • Latin American community HK	• Women Founders HK • AI Community HK • FinTech Association HK (HKFA) • RISE Conference network
STUDENT COMMUNITIES • University clubs by nationality • International student networks • Alumni associations HK • Entrepreneurship clubs at HKU/HKUST • Research collaboration groups	REAL SOCIAL INTEGRATION • Running clubs & sports groups • Volunteering (HKTE Volunteer) • Cultural & language groups • Networking social events • "M'goi" — the word that opens doors

6. HOW IT WORKS — CITYVOICE

The user accesses everything above through a single voice interaction. No forms. No English required. Just talking.

- 1 They speak in their language — INPUT**
 Spanish, Mandarin, German, English, Hindi, any language. Natural. Like talking to someone who already knows HK.
- 2 Claude understands their profile — AI**
 Extracts who they are, what they want, where they're from, what stage they're at. Without asking explicitly — inferred from how they speak.
- 3 The constellation map appears (nodes) — OUTPUT**
 Possibilities + communities + next actions. All personalised. All real. Star by star, the constellation builds as they speak.

7. THE DEMO SCRIPT — 5 MINUTES ON STAGE

Stage setup

Left screen: live chat — transcribed text visible + Claude's response appearing in real time.

Right screen: empty constellation map at the start. Stars appear live as Claude speaks. Dashed lines connect them like a star map building in real time.

8. WHY THIS IS SMART CITY

IF THE JURY ASKS — SAY THIS EXACTLY

Smart City no longer means only sensors and traffic. Modern smart cities compete for human capital. A city that loses 40% of its international talent in the first year is not smart — regardless of its physical infrastructure. City Twin is the missing layer.

SMART CITY 1.0 — WHAT EVERYONE ELSE BUILDS • Sensors and IoT dashboards • Traffic and mobility optimisation • Energy efficiency tools • Municipal data visualisations • Infrastructure monitoring	SMART CITY 2.0 — WHAT CITY TWIN BUILDS • Human integration infrastructure • Talent retention and attraction • City intelligence for human capital • Personalised urban navigation in any language • The layer that makes people stay
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9. TECHNICAL STACK — REALISTIC 48 HOURS

VOICE	Web Speech API Native in Chrome. Free. Real-time transcription in any language. No installation.	1–2h
LANGUAGE	Claude API Auto-detects language inside the same prompt. No extra library needed.	30 min
AI	Claude API + system prompt Extracts profile from spoken text. Forces English response. Returns nodes as structured JSON.	3–4h
MAP	Canvas API (constellation) Animated star map already built. Connect nodes from Claude's JSON to the constellation canvas.	3–4h
SYNC	Timestamps + keywords When Claude returns a node, a star appears. Progressive with streaming.	2–3h
UI	React + Tailwind Chat left + constellation right. Dark mode. Deployed on Vercel. Mobile-first for stage.	3–4h
BACKUP	3 pre-recorded videos One per profile (Sofia/Lena/Priya), perfectly synced. Invisible fallback if live fails.	2h

THE CORE CLAUDE PROMPT

You are City Twin. The user speaks in any language. Detect their language. Extract their profile (role, field, challenge, time in HK). ALWAYS respond in English. Mention their

original language warmly. Generate 3-5 specific HK ecosystem nodes as JSON: [{name, category, reason}]. Only real HK organisations. Never be generic.

Priority order if time runs short

1. English voice + working constellation end-to-end. This alone is already competitive.
2. English or German demo working. The single most impactful language moment for HKTE jury.
3. All three profiles (Sofia, Lena, Priya) demonstrated. Only if P1 and P2 are solid.
4. Constellation animation polish + streaming sync. Nice to have. Do not sacrifice working demos for visual effects.

10. TEAM DIVISION — 4 PEOPLE, 48 HOURS (DRAFT)

Dev 1 — Frontend + Map	• UI: chat left + constellation right • Connect JSON nodes to animated canvas • Star entrance animations + dashed lines • Deploy on Vercel + final visual polish
Dev 2 — AI + Voice	• Web Speech API + language detection • Claude API integration • Multilingual prompt engineering • Structured JSON output + streaming
Dev 3 — Data + Backup	• 50+ real HK opportunities in JSON • Record 3 backup videos (Sofia/Lena/Priya) • Sync backups with UI • Test with Spanish, German and Indian English accents
María — Pitch + Strategy	• pitch slides (problem, solution, team) • Coordinate demo and design • City Twin narrative + real traction • Business and Marketing

What we do NOT build

Simplicity executed to perfection always beats half-finished ambition.

- ❌ AR / Augmented reality
- ❌ Dynamic database with auth
- ❌ Login or user accounts
- ❌ Native mobile app
- ❌ More than one user flow
- ❌ Complex backend infrastructure

11. RISKS AND MITIGATIONS

20%	Web Speech API fails with accents Mitigation: Pre-recorded video per profile (Sofia/Lena/Priya). If live fails, run backup — jury won't notice the difference.
10%	Claude responds in the wrong language Mitigation: System prompt forces English response always. Tested before the stage with all 3 demo profiles and languages.
15%	Constellation builds out of sync Mitigation: Hardcoded timestamps for demo keywords. If streaming fails, stars appear in fixed order. Still impactful.
5%	Hackathon wifi slow or down Mitigation: Full demo works offline with hardcoded data + backup videos. No network dependency for the main demo.
10%	Build time runs out Mitigation: Priority 1: English voice + working constellation. Non-English profiles are bonuses. English-only is already competitive.

THE TRUTH ABOUT BACKUPS

Backups are not Plan B. They are Plan A-prime. Record all 3 profile videos (Sofia, Lena, Priya) with the same quality as the live demo. Rehearse them as much as the real thing.

12. 48-HOUR EXECUTION CHECKLIST

Day 1 · First 12 hours — Core build

- ☐ Web Speech API working with real-time transcription
- ☐ Claude API integrated with multilingual system prompt
- ☐ JSON node output working correctly
- ☐ Constellation map connected to Claude JSON
- ☐ Basic UI: chat left + constellation right
- ☐ English demo working end-to-end

Day 1 · Next 12 hours — Three profiles + polish

- ☐ Sofia demo working (Spanish — student, Madrid, not yet in HK)

- ☐ Lena demo working (German — entrepreneur, 1 month in HK, AI idea)
- ☐ Priya demo working (English — AI researcher from India, just arrived)
- ☐ Star animations polished + dashed lines visible
- ☐ Deployed on Vercel

Day 2 · Final 24 hours — Backup + rehearsal

- ☐ Backup video recorded: Sofia (Spanish profile)
- ☐ Backup video recorded: Lena (German profile)
- ☐ Backup video recorded: Priya (English / Hindi profile)
- ☐ Tested.
- ☐ Pitch slides ready
- ☐ Full rehearsal

The 3 sentences to close every conversation about this project.

01

"A city is not truly smart if the people inside it feel lost."

02

"Connecting global talent to HK — before they give up, and before they even arrive."

03

"We are not asking for funding. We are asking for 100 users."

Good luck in Munich.

City Twin · EuroTech x HKTE Hackathon · Smart City / AI & Robotics